

1. The SI unit of permeability of free space is

- a) $\frac{\text{weber}}{\text{ampere}}$ b) $\frac{\text{henry}}{\text{ampere}}$ c) $\frac{\text{tesla}}{\text{ampere} - \text{meter}}$ d) $\frac{\text{weber}}{\text{ampere} - \text{meter}}$

Sol Permeability of free space: μ_0

Magnetic field due to straight current carrying infinitely long conductor: $B = \frac{\mu_0 i}{2\pi r} \rightarrow \mu_0 = \frac{2\pi r B}{i}$

SI unit of magnetic field induction (flux density): B (weber/m²)

SI unit of current: i (ampere)

SI unit of distance: r (meter)

So the SI unit of permeability of free space: $\frac{\text{meter} \times \frac{\text{weber}}{\text{meter}^2}}{\text{ampere}} = \frac{\text{weber}}{\text{ampere} - \text{meter}}$

2. A uniform solid drum of radius R and mass M rolls without slipping down a plane inclined at an angle θ . Its acceleration along the plane is

- a) $\frac{1}{3} g \sin \theta$ b) $\frac{1}{2} g \sin \theta$ c) $\frac{2}{3} g \sin \theta$ d) $\frac{5}{7} g \sin \theta$

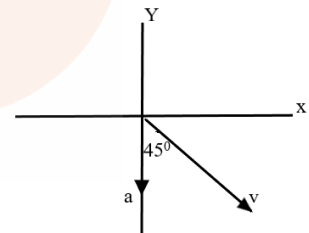
Sol Acceleration of a rolling object down the inclined plane: $a = \frac{g \sin \theta}{1 + \frac{K^2}{R^2}}$

For solid drum (solid cylinder): $\frac{K^2}{R^2} = \frac{1}{2}$

$$a = \frac{g \sin \theta}{1 + \frac{K^2}{R^2}} = \frac{g \sin \theta}{1 + \frac{1}{2}} = \frac{2}{3} g \sin \theta$$

3. A particle moves according to the law $x = at$, $y = at(1 - \alpha t)$ where a and α are positive constants and t is time. The instant at which velocity makes an angle $\frac{\pi}{4}$ with acceleration is

- a) $\frac{4}{\alpha}$ b) $\frac{3}{\alpha}$ c) $\frac{2}{\alpha}$ d) $\frac{1}{\alpha}$



Sol Instantaneous position of the particle along x axis: $x = at$

Instantaneous position of the particle along y axis: $y = at(1 - \alpha t)$

Velocity of the particle: $\vec{v} = v_x \hat{i} + v_y \hat{j} = a \hat{i} - (a - 2\alpha at) \hat{j}$

Acceleration of the particle: $\vec{a} = a_x \hat{i} + a_y \hat{j} = 0 \hat{i} - 2\alpha \hat{j}$

Given: angle made by velocity vector with acceleration vector is 45°

So, the angle made by velocity vector with X axis is also 45°

$$\tan(-45) = \frac{v_y}{v_x} \rightarrow -1 = \frac{a(1 - 2\alpha t)}{a} \rightarrow -a = a(1 - 2\alpha t) \rightarrow 2\alpha t = 2 \rightarrow t = \frac{1}{\alpha}$$

4. The potential energy of a mass m in a conservative force field can be expressed as $U = \alpha x - \beta y$ where (x, y) denote the position coordinates of the body. The acceleration of the body is

- a) $\frac{\alpha - \beta}{m}$ b) $\frac{\alpha + \beta}{m}$ c) $\frac{\sqrt{\alpha^2 - \beta^2}}{m}$ d) $\frac{\sqrt{\alpha^2 + \beta^2}}{m}$

Sol Potential energy of the particle: $U = \alpha x - \beta y$
 force acting on the particle: $F = -\frac{dU}{dr}$ for a single variable function
 Force acting on the particle: $\vec{F} = -\frac{\partial U}{\partial x} \hat{i} - \frac{\partial U}{\partial y} \hat{j} = -\alpha \hat{i} + \beta \hat{j}$
 Acceleration of the particle: $\vec{a} = \frac{\vec{F}}{m} = \frac{1}{m} (-\alpha \hat{i} + \beta \hat{j})$
 Magnitude of acceleration: $a = \frac{\sqrt{\alpha^2 + \beta^2}}{m}$

5. A constant force F applied to the lower block of mass 15 kg makes it slide between the upper block of mass 5 kg and the table below, as shown. The coefficient of static (μ_s) and kinetic (μ_k) friction between the lower block and the table are 0.5 and 0.4 respectively and those between the two blocks are 0.3 and 0.1. The acceleration of the upper and the lower blocks are respectively
- a) 1.96 m/s^2 and 1.96 m/s^2 b) 1.96 m/s^2 and 3.92 m/s^2
 c) 0.98 m/s^2 and 0.49 m/s^2 d) **0.98 m/s^2 and 1.96 m/s^2**

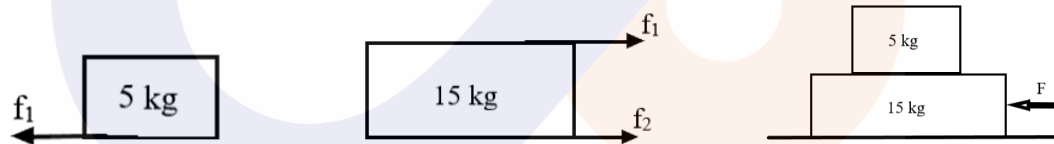
Sol Force required to slide the 15 kg block:

$$F = \mu_{s1}mg + \mu_{s2}(m + M)g = 0.3 \times 5 \times 9.8 + 0.5 \times 20 \times 9.8 = 112.7 \text{ N}$$

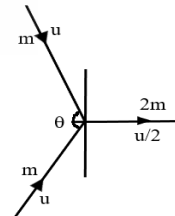
$$\text{For the 5 kg block: } f_1 = ma_1 \rightarrow \mu_{k1}mg = ma_1 \rightarrow a_1 = \mu_{k1}g = 0.1 \times 9.8 = \mathbf{0.98 \text{ m/s}^2}$$

$$\text{For the 15 kg block: } F - f_1 - f_2 = Ma_2 \rightarrow F - \mu_{k1}mg - \mu_{k2}(m + M)g = Ma_2$$

$$112.7 - 0.1 \times 5 \times 9.8 - 0.4 \times 20 \times 9.8 = 15 a_2 \rightarrow 112.7 - 4.9 - 78.4 = 15 a_2 \rightarrow \mathbf{a_2 = 1.96 \text{ m/s}^2}$$



6. Two bodies of equal masses moving with equal speeds make a perfectly inelastic collision. If the speed after the collision is reduced to half, the angle between their velocities of approach is



- a) 30° b) 60° c) 90° d) **120°**

Sol Masses of the two bodies are equal: $m_1 = m_2 = m$

Speeds of the two bodies before collision are equal: $u_1 = u_2 = u$

Let the angle between the velocities of approach be θ

$$\text{Initial momentum: } P_i = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta} = \sqrt{(mu)^2 + (mu)^2 + 2(mu)(mu) \cos \theta}$$

$$\text{Final momentum of the system: } P_f = 2m \frac{u}{2}$$

$$\text{Conservation of momentum: } P_f = P_i \rightarrow m^2u^2 = 2m^2u^2 + 2m^2u^2 \cos \theta \rightarrow -m^2u^2 = 2m^2u^2 \cos \theta$$

$$\cos \theta = -\frac{1}{2} \rightarrow \mathbf{\theta = 120^\circ}$$

7. A student performs an experiment with a simple pendulum and records the time for 20 oscillations. If he would have recorded time for 100 oscillation, the error in the measurement of time period would have reduced by a factor of

a) 80 b) 20 c) 10 d) **5**

Sol Let the measured values: $a_1, a_2, a_3, \dots, a_n$

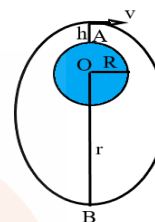
$$\text{Mean value: } a_{\text{mean}} = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$$

$$\text{Absolute error in the measurement: } \Delta a_1 = |a_1 - a_{\text{mean}}|$$

$$\text{Mean absolute error: } \Delta a_{\text{mean}} = \frac{\Delta a_1 + \Delta a_2 + \Delta a_3 + \dots + \Delta a_n}{n}$$

From the above expression, mean absolute error is inversely related to number of observations. As the number of observations increased by a factor of 5, the error in the measurement of time period would reduce by a factor of **5**

8. A satellite is launched from a point close to the surface of the earth (radius R) with velocity $v = v_0 \sqrt{1.5}$, where v_0 is the velocity in a circular orbit. If the initial velocity imparted to the satellite is horizontal, the maximum distance from the surface of the earth during its revolutions is



a) R b) **$2R$** c) $3R$ d) $4R$

Sol Orbital velocity of a satellite: $v_0 = \sqrt{\frac{GM}{r}}$

The satellite follows an elliptical path with center of the earth as one of focal points as shown.

$$\text{Conservation of angular momentum: } L_i = L_f \rightarrow mvR = mVr \rightarrow v_0 \sqrt{1.5} R = Vr \rightarrow V = \frac{v_0 \sqrt{1.5} R}{r}$$

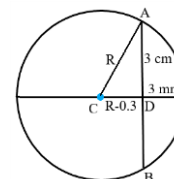
$$\text{Conservation of energy: } E_A = E_B \rightarrow -\frac{GM}{R} + \frac{1}{2}m(1.5)v_0^2 = -\frac{GM}{r} + \frac{1}{2}mV^2$$

$$-\frac{1}{4} \frac{GM}{R} = -\frac{GM}{r} + \frac{3}{4} \frac{GM}{r^2} R \rightarrow -\frac{1}{4R} = -\frac{1}{r} + \frac{3R}{4r^2} \rightarrow r^2 - 4Rr + 3R^2 = 0 \rightarrow r = R \text{ or } 3R$$

The maximum distance of the satellite from the center of the earth is $3R$

The maximum distance of the satellite from the surface of the earth is **$2R$**

9. The aperture diameter of a plano-convex lens is 6 cm and its thickness is 3 mm. If the speed of light through its material is $v = 2 \times 10^8$ m/s, the focal length of the lens is



a) 40 cm b) 35 cm c) **30 cm** d) 20 cm

Sol Aperture diameter of the plano-convex lens: $AB = 6$ cm

Thickness of the plano-convex lens: $3 \text{ mm} = 0.3 \text{ cm}$

Speed of light through the material of the lens: $v = 2 \times 10^8$ m/s

$$\text{Refractive index of the material: } n = \frac{c}{v} = \frac{3 \times 10^8}{2 \times 10^8} = 1.5$$

$$\text{From } \Delta^{\text{le}} \text{ CAD: } R^2 = 9 + (R-0.3)^2 \rightarrow R = 15 \text{ cm}$$

$$\text{Lens maker's formula: } \frac{1}{f} = (n-1) \frac{1}{R} \rightarrow \frac{1}{f} = (1.5-1) \frac{1}{15} = \frac{1}{30} \rightarrow \mathbf{f = 30 \text{ cm}}$$

10. Under standard conditions of temperature and pressure a piece of ice melts completely on heating it. obviously, the increase in internal energy of the system (ice and water) is

a) equal to the heat given
 b) **more than the heat given**
 c) less than heat given
 d) zero

Sol first law of thermodynamics: $dQ = dU + dW$

dQ – heat energy: supplied to the system (+ve) and removed from the system (–ve)

dW – work done: by the system (+ve) and on the system (–ve)

dU – change in internal energy: increase (+ve) and decrease (–ve)

As ice melts, its volume decreases: $dW = P dV = -P dV$

$$dU = dQ - dW = \mathbf{dQ + P dV}$$

So, the increase in internal energy of the system (ice + water) is more than the heat given.

11. Rocket fuel is capable of giving an exhaust velocity of $v_{rel} = 2.4$ km/s in the absence of any external forces. The fuel required per kg of the payload to provide an exhaust velocity of 12 km/s to the rocket is

a) 3670 kg b) 8000 kg c) **147.4 kg** d) 478.4 kg

Sol Instantaneous speed of the rocket: $v = v_{rel} \ln \frac{m_0}{m}$

m_0 – initial mass of rocket: payload (satellite) + fuel

m – instantaneous mass of the rocket: payload(satellite) + remaining fuel

$$12 = 2.4 \ln \frac{m_0}{m} \rightarrow 5 = \ln \frac{m_0}{m} \rightarrow \frac{m_0}{m} = e^5 \rightarrow m_0 = m e^5$$

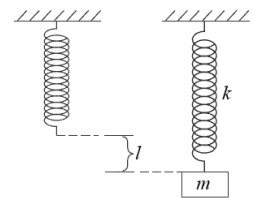
$$\text{Mass of the fuel required: } m_0 - m = m e^5 - m = m (e^5 - 1) = 1 (148.4 - 1) = \mathbf{147.4 \text{ kg}}$$

12. A vertical spring of length l_0 and force constant k is stretched by l when a mass m is suspended from its lower end. By pulling the mass down a little the system is left off to oscillate. The time period of oscillation is

a) **$2\pi \sqrt{\frac{l}{g}}$** b) $2\pi \sqrt{\frac{l_0}{g}}$ c) $\frac{1}{2\pi} \sqrt{\frac{m}{k}}$ d) $2\pi \sqrt{\frac{l+l_0}{g}}$

Sol Equilibrium condition: $mg = kl \rightarrow k = \frac{mg}{l}$

$$\text{Time period of spring block oscillations: } T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{m}{\frac{mg}{l}}} = \mathbf{2\pi \sqrt{\frac{l}{g}}}$$



13. Let R be the radius of the earth. In general, the loss of gravitational potential energy of a body of mass m falling from a height h to the earth surface is

a) mgh b) **$mgh \frac{R}{R+h}$** c) $mgh \sqrt{\frac{R+h}{R}}$ d) $mgh \sqrt{\frac{R}{R+h}}$

Sol Potential energy of the object on the earth: $U_1 = -\frac{GMm}{R}$
 Potential energy of the object at a height h : $U_2 = -\frac{GMm}{R+h}$
 Loss of potential energy: $\Delta U = U_2 - U_1 = -\frac{GMm}{R+h} + \frac{GMm}{R} = GMm \left(\frac{1}{R} - \frac{1}{R+h} \right)$
 $\Delta U = GMm \frac{h}{R(R+h)} = \frac{GM}{R^2} \frac{mh}{1+\frac{h}{R}} = \frac{mgh}{1+\frac{h}{R}} = mgh \frac{R}{R+h}$

14. The velocity of a projectile at the highest point of its trajectory is $\sqrt{0.4}$ of its velocity at a point at half its maximum height. The angle of projection is

a) 30° b) 45° c) 60° d) $\tan^{-1}(\sqrt{0.4})$

Sol Initial velocity of the projectile: u and Velocity at the highest point: $u \cos \theta$

Velocity at half of maximum height: $v = \sqrt{(u \cos \theta)^2 + \frac{1}{2}(u \sin \theta)^2}$

Given: $u^2 \cos^2 \theta = (0.4) \left(u^2 \cos^2 \theta + \frac{u^2 \sin^2 \theta}{2} \right)$

$0.6 u^2 \cos^2 \theta = 0.2 u^2 \sin^2 \theta \rightarrow \tan^2 \theta = 3 \rightarrow \tan \theta = \sqrt{3} \rightarrow \theta = 60^\circ$

15. The combination of a steel wire (length 80 cm, area of cross section 1 mm^2) and an aluminium wire (length 60 cm, area of cross section 3 mm^2) joined end to end is stretched by a tension of 160 N. If the densities of steel and aluminum are 7.8 g/cc and 2.6 g/cc respectively then, the minimum frequency of a tuning fork which can produce standing waves in the composite wire, with the joint as a node, is

a) 170 Hz b) **358 Hz** c) 88 Hz d) 118 Hz

Sol Frequency of transverse vibrations of a stretched string: $f = \frac{n}{2l} \sqrt{\frac{T}{\mu}} = \frac{n}{2l} \sqrt{\frac{T}{A\rho}}$

Frequency of oscillation of steel wire: $f_{st} = \frac{n_{st}}{2l_{st}} \sqrt{\frac{T_{st}}{A_{st} \rho_{st}}} = \frac{n_{st}}{2 \times 80} \sqrt{\frac{160}{1 \times 7.8}} = \frac{n_{st}}{160} \sqrt{\frac{160}{7.8}}$

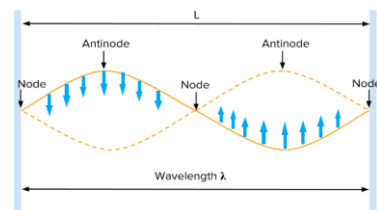
Frequency of oscillation of aluminum wire: $f_{al} = \frac{n_{al}}{2l_{al}} \sqrt{\frac{T_{al}}{A_{al} \rho_{al}}} = \frac{n_{al}}{2 \times 60} \sqrt{\frac{160}{3 \times 2.6}} = \frac{n_{al}}{120} \sqrt{\frac{160}{7.8}}$

In order to have a node at the joint: $f_{st} = f_{al} \rightarrow \frac{n_{st}}{160} \sqrt{\frac{160}{7.8}} = \frac{n_{al}}{120} \sqrt{\frac{160}{7.8}} \rightarrow \frac{n_{st}}{n_{al}} = \frac{4}{3}$

So, frequency of the tuning fork: $f_{st} = \frac{4}{2 \times 0.8} \sqrt{\frac{160}{1 \times 10^{-6} \times 7.8 \times 10^3}} = 2.5 \times 143.23 = \mathbf{358 \text{ Hz}}$

16. In a stationary wave

- a) all the medium particles vibrate in the same phase
 b) **all the particles between two consecutive nodes vibrate in the same phase**
 c) any two consecutive nodes vibrate in the same phase
 d) all the particles between two consecutive antinodes vibrate in the same phase



Sol Node: the point of minimum (zero) displacement.

Antinode: the point of maximum displacement.

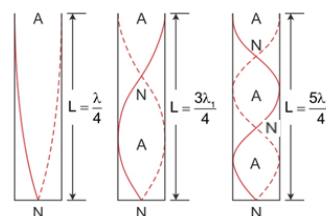
Once a stationary wave is formed between two fixed points, transfer of energy does not take place.

Energy is not transferred between one node and the other, instead it gets transformed between kinetic energy and elastic potential energy.

At a given instant, **all the particles between two nodes will be moving in same fashion with same frequency, time period and phase** so that the shape of the wave is sustained.

17. An empty earthen pitcher is kept under a water tap and starts filling water as the tap is opened. The pitch of the sound produced

- a) goes on decreasing
- b) **goes on increasing**
- c) first increases and then decreases after the pitcher is half filled
- d) does not change



Sol The empty earthen pitcher can be approximated to a closed organ pipe.

Frequency of oscillation of the air column: **$f = \frac{(2n+1)v}{4l}$** (only odd harmonics are present)

As water is filled into the pitcher, the length of the air column in it decreases, thus increasing the frequency.

18. The molar specific heat of an ideal gas in a certain thermodynamic process is $\frac{\alpha}{T}$ where α is a constant. If the adiabatic exponent is $\gamma = \frac{c_p}{c_v}$, the work done in heating the gas from T_0 to nT_0 is

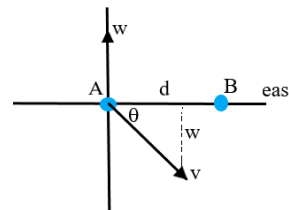
- a) $\frac{1}{\alpha} \ln n$
- b) **$\alpha \ln n - \left(\frac{n-1}{\gamma-1}\right) RT_0$**
- c) $\alpha \ln n - (\gamma - 1) RT_0$
- d) $\left(\frac{n-1}{\gamma-1}\right) RT_0$

Sol Heat supplied to the ideal gas: $dQ = \int_{T_0}^{nT_0} c dT = \int_{T_0}^{nT_0} \frac{\alpha}{T} dT = \alpha (\ln nT_0 - \ln T_0) = \alpha \ln n$

Change in internal energy: $dU = -dW = \frac{R}{\gamma-1} (nT_0 - T_0) = RT_0 \left(\frac{n-1}{\gamma-1}\right)$

First law of thermodynamics: $dQ = dU + dW \rightarrow dW = dQ - dU = \alpha \ln n - RT_0 \left(\frac{n-1}{\gamma-1}\right)$

19. An aircraft flies at a speed v from city A to city B and back in time t_0 . City B is to the east of city A at a distance d . The aircraft takes time t_1 for the round trip if wind blows with speed w along AB and time t_2 if the wind blows with the same speed perpendicular to AB. Then,



- a) $t_1 = t_2 = t_0$
- b) **$t_1 > t_2 > t_0$**
- c) $t_1 < t_2 < t_0$
- d) $t_1 > t_0 > t_2$

Sol Distance between the two cities: d

Time taken by the aircraft for round trip: $t_0 = \frac{2d}{v}$

Time taken by the aircraft for round trip when wind blows along AB: $t_1 = \frac{d}{v+w} + \frac{d}{v-w} = \frac{2vd}{v^2-w^2}$

Time taken by the aircraft for round trip when wind blows perpendicular to AB:

$$t_2 = \frac{2d}{v \cos \theta} = \frac{2d}{v \frac{\sqrt{v^2-w^2}}{v}} = \frac{2d}{\sqrt{v^2-w^2}}$$

20. The Hubble telescope in orbit above the earth has a 2.4 m circular aperture. The telescope has equipment for detecting ultraviolet light. The minimum angular separation between two objects that the telescope can resolve in ultraviolet light of wavelength 95 nm is

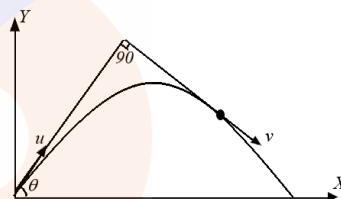
- a) $4.83 \times 10^{-8} \text{ rad}$ b) $4.03 \times 10^{-8} \text{ rad}$ c) $2.41 \times 10^{-8} \text{ rad}$ d) $2.00 \times 10^{-8} \text{ rad}$

Sol Circular aperture of the telescope: $d = 2.4 \text{ m}$

Wavelength of the ultraviolet light: $\lambda = 95 \text{ nm}$

The minimum angular separation between two objects: $\frac{1.22\lambda}{d} = \frac{1.22 \times 95 \times 10^{-9}}{2.4} = 4.83 \times 10^{-8} \text{ rad}$

21. A projectile is fired from ground with velocity u at an angle θ with the horizontal. It would be moving perpendicular to its initial direction of projection after a time t equal to



- a) $\frac{u \sin \theta}{g}$ b) $\frac{2u \sin \theta}{g}$ c) $\frac{u}{g \sin \theta}$ d) $\frac{u}{2g \sin \theta}$

Sol Initial velocity of the projectile: $\vec{u} = u \cos \theta \hat{i} + u \sin \theta \hat{j}$

Velocity of the projectile at time t : $\vec{v} = u \cos \theta \hat{i} + (u \sin \theta - gt) \hat{j}$

Given: $\vec{u} \cdot \vec{v} = 0 \rightarrow (u \cos \theta \hat{i} + u \sin \theta \hat{j}) \cdot (u \cos \theta \hat{i} + (u \sin \theta - gt) \hat{j}) = 0$

$$u^2 \cos^2 \theta + u^2 \sin^2 \theta - u \sin \theta (gt) = 0 \rightarrow u^2 = u \sin \theta (gt) \rightarrow t = \frac{u}{g \sin \theta}$$

22. The critical angle for light passing from glass to air is minimum for the light of wavelength

- a) $0.7 \mu\text{m}$ b) $0.6 \mu\text{m}$ c) $0.5 \mu\text{m}$ d) $0.4 \mu\text{m}$

Sol Critical angle for glass to air: $\sin c = \frac{1}{\mu} = \frac{\lambda}{\lambda_0}$

λ_0 - wavelength of light in vacuum AND λ - wavelength of light in glass

$$\sin c = \frac{\lambda}{\lambda_0} \rightarrow \sin c \propto \lambda \text{ [The minimum value of wavelength from the given options: } 0.4 \mu\text{m}]$$

23. A thin hollow equiconvex lens, silvered at the back, converges a beam of light parallel to the principal axis at a distance 0.2 m, when filled with water ($\mu = \frac{4}{3}$), the same beam will be converged at a distance of

a) 0.4 m b) 0.20 m c) **0.12 m** d) none the above

Sol Since the incoming beam of light is parallel to principal axis, it converges at focal point: $f = 0.2$ m
A thin hollow equiconvex lens, silvered at the back behaves like a concave mirror: $R = 2f = 0.4$ m
When water is filled, the focal length of the water lens:

$$\frac{1}{f_w} = (n_w - 1) \frac{2}{R} \rightarrow \frac{1}{f_w} = \left(\frac{4}{3} - 1\right) \frac{2}{0.4} \rightarrow f_w = 0.6 \text{ m}$$

$$\text{Effective focal length of the system: } \frac{1}{F} = \frac{2}{f_w} + \frac{1}{f} = \frac{2}{0.6} + \frac{1}{0.2} \rightarrow \mathbf{F = 0.12 \text{ m}}$$

24. An air bubble is situated at a distance 2.0 cm from the Centre of a spherical glass paper-weight of radius 5.0 cm and refractive index 1.5. The bubble is seen through the nearest surface. It appears at a distance v from the Centre. Therefore, v is

a) 3.75 cm b) 3.25 cm c) **2.50 cm** d) 3.80 cm

Sol refraction of light at curved surface: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$

n_1 – RI of the medium in which incident ray is present = 1.5

n_2 – RI of the medium in which refracted ray is present = 1

$u = 3$ cm; $R = 5$ cm;

$$\frac{1}{v} + \frac{1.5}{3} = \frac{1-1.5}{-5} \rightarrow \frac{1}{v} = -0.5 + 0.1 = -0.4 \rightarrow v = \frac{1}{0.4} = 2.5 \text{ cm (from the surface)}$$

Distance of the image from the center: $d = 5 - v = 5 - 2.5 = \mathbf{2.5 \text{ cm}}$

25. student while performing experiment with a sonometer with bridge separated by distance of 80 cm, missed out some of the observations. However, he claimed that the three resonant frequencies for a given tuning fork were 84, 140 and 224 Hz. The speed transverse waves on the wire is

a) 33.30 m/s b) 330.0 m/s c) **44.80 m/s** d) 448.0 m/s

Sol Separation between the bridges: $l = 80$ cm = 0.8 m

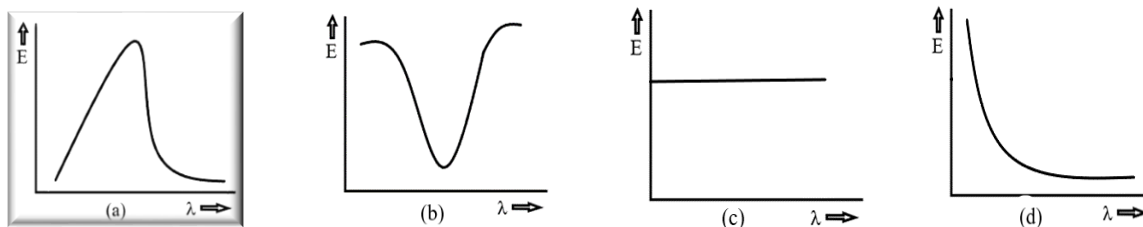
Resonant frequencies of the tuning fork: 84, 140, 224

The given frequencies are in the ratio: 3: 5: 8

$$\text{Frequency of transverse wave on a stretched string: } f = \frac{n}{2l} \sqrt{\frac{T}{\mu}} = \frac{nv}{2l}$$

$$84 = \frac{3v}{2 \times 0.8} \rightarrow \mathbf{v = 44.80 \text{ m/s}}$$

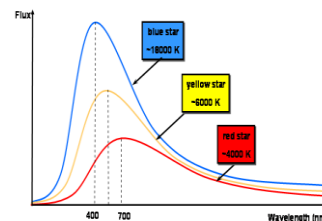
26. Which of the following curves represents spectral distribution of energy of black body radiation?



Sol Planck's black body radiation law: $E_\lambda = \frac{8\pi hc}{\lambda^5 \left(e^{\frac{hc}{\lambda kT}} - 1 \right)}$

Studies of black body radiation by Planck laid the foundation for the concept of photon, which was later firmly established by Einstein while explaining the observations of photo electric effect.

From the diagram, the correct option is (a)



27. sphere and a cube having equal surface area are made of the same material. The two are heated to the same temperature and kept in identical surroundings. The ratio of their initial rates of cooling is

- a) 1:1 b) $\sqrt{\frac{\pi}{2}} : 1$ c) $\sqrt{\frac{\pi}{3}} : 1$ d) $\sqrt{\frac{\pi}{6}} : 1$

Sol Surface area of sphere = surface area of cube $\rightarrow 4\pi r^2 = 6a^2 \rightarrow \frac{r}{a} = \sqrt{\frac{3}{2\pi}}$

The two objects are made of same material: density and specific heat capacity are same

Newton's law of cooling: $\frac{dQ}{dt} = -b(T - T_0)$

$$ms \frac{dT}{dt} = -b(T - T_0) \rightarrow \frac{dT}{dt} = -\frac{b}{ms}(T - T_0) = -\frac{b}{v\rho s}(T - T_0) \rightarrow \frac{dT}{dt} \propto \frac{1}{v}$$

$$\text{Ratio of initial rates of cooling: } \frac{v_c}{v_s} = \frac{a^3}{\frac{4}{3}\pi r^3} = \frac{3}{4\pi} \left(\frac{a}{r}\right)^3 = \frac{3}{4\pi} \left(\frac{2\pi}{3}\right)^{3/2} = \sqrt{\frac{\pi}{6}} : 1$$

28. Consider the diffraction pattern due to a single slit. The first maximum for a certain monochromatic light coincides with the first minimum for red light of wavelength 660 nm. The wavelength of the monochromatic light is

- a) 660 nm b) 550 nm c) 440 nm d) 330 nm

Sol Condition for diffraction minima:

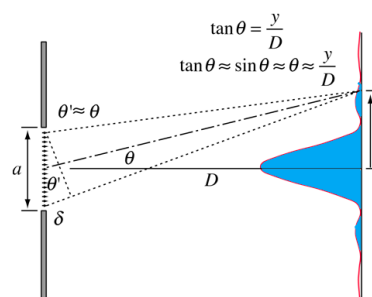
$$a \sin \theta = n\lambda \rightarrow a \frac{y}{D} = \lambda_r \rightarrow y = \frac{D}{a} \lambda_r \text{ ---- (1)}$$

(a: slit width, n is order of the minima, $\sin \theta = y/D$)

Condition for diffraction maxima:

$$a \sin \theta = \left(n + \frac{1}{2}\right) \lambda \rightarrow a \frac{y}{D} = \frac{3}{2} \lambda \rightarrow y = \frac{3}{2} \frac{D}{a} \lambda \text{ ---- (2)}$$

$$\text{From (1) and (2): } \frac{D}{a} \lambda_r = \frac{3}{2} \frac{D}{a} \lambda \rightarrow \lambda = \frac{2}{3} \lambda_r = \frac{2}{3} \times 660 = 440 \text{ nm}$$



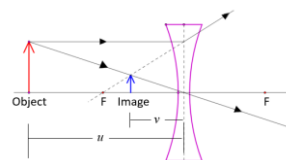
29. A concave lens of focal length f produces an image $(1/n)$ times the size of the object. The distance of the object from the lens is

a) $(n+1)f$ b) $\frac{(n-1)f}{n}$ c) $\frac{(n+1)f}{n}$ d) **$(n-1)f$**

Sol Size of the image = $\frac{1}{n}$ size of the object $\rightarrow h_i = \frac{1}{n} h_o \rightarrow \frac{h_i}{h_o} = \frac{1}{n} = m$

Magnification in terms of distances: $m = \frac{v}{u} \rightarrow \frac{1}{n} = \frac{v}{u} \rightarrow v = \frac{u}{n}$

For a concave lens: u, v and f are $-ve$



Lens formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \rightarrow -\frac{1}{v} + \frac{1}{u} = -\frac{1}{f} \rightarrow -\frac{n}{u} + \frac{1}{u} = -\frac{1}{f} \rightarrow \frac{n-1}{u} = \frac{1}{f} \rightarrow u = (n-1)f$

30. The sun having radius R and surface temperature T , emits radiation as a perfect emitter. The distance of the earth from the sun is r and the radius of the earth is R_e . The total radiant power incident on the earth is

a) $\frac{R_e^2 R^2 \sigma T^4}{4\pi r^2}$ b) $\frac{R_e^2 R^2 \sigma T^4}{r^2}$ c) $\frac{4\pi R_e^2 R^2 \sigma T^4}{r^2}$ d) **$\frac{\pi R_e^2 R^2 \sigma T^4}{r^2}$**

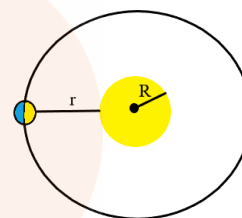
Sol Intensity of radiation emitted by sun: $I_s = \sigma T^4$

Surface area of sun: $A_s = 4\pi R^2$

Power radiated by sun: $P_s = I_s A_s = \sigma T^4 (4\pi R^2)$

Projected area of earth facing sun: πR_e^2

Power received by earth: $P_e = \frac{A_e}{A} P_s = \frac{\pi R_e^2}{4\pi r^2} \sigma T^4 (4\pi R^2) = \frac{\pi R_e^2 R^2 \sigma T^4}{r^2}$



31. cylinder containing water (refractive index $4/3$) is covered by an equiconvex glass (refractive index $3/2$) lens of focal length 25 cm. At the mid-day when the sun is just overhead, the image of the sun will be seen at a distance of

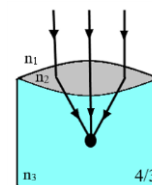
a) 100 cm b) **50 cm** c) 37.5 cm d) 25 cm

Sol Lens maker's formula for equi-convex lens:

$\frac{1}{f} = (n-1) \frac{2}{R} \rightarrow \frac{1}{25} = (1.5-1) \frac{2}{R} \rightarrow R = 25 \text{ cm}$

Modified form of lens maker's formula for different media on either side

$\frac{1}{f} = \frac{1}{n_3} \left[\frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2} \right] \rightarrow \frac{1}{f} = \frac{3}{4} \left[\frac{\frac{3}{2} - 1}{25} + \frac{\frac{4}{3} - \frac{3}{2}}{-25} \right] \rightarrow f = 50 \text{ cm}$



32. A rectangular loop carrying a current is placed in a uniform magnetic field. The net force acting on the loop

- a) depends on the direction and magnitude of the current
b) depends on the direction and magnitude of the magnetic field
c) depends on the area of the loop
d) **is zero**

Sol Let the magnetic field induction: B

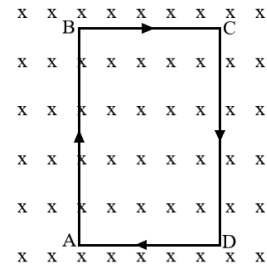
Force on the conductor AB: $F_{AB} = B i l$ (left)

Force on the conductor BC: $F_{BC} = B i b$ (up)

Force on the conductor CD: $F_{CD} = B i l$ (right)

Force on the conductor DA: $F_{DA} = B i b$ (down)

So, the net force on the rectangular loop: $\mathbf{F} = \mathbf{F}_{AB} - \mathbf{F}_{CD} + \mathbf{F}_{BC} - \mathbf{F}_{DA} = \mathbf{0}$



33. The capacitor in the circuit shown below carries a charge of $30 \mu\text{C}$ at a certain time instant. The rate at which energy is being dissipated in the $3 \text{ k}\Omega$ resistor at that instant is

- a) 4 mW b) 9 mW c) **27 mW** d) 48 mW

Sol Instantaneous charge on capacitor:

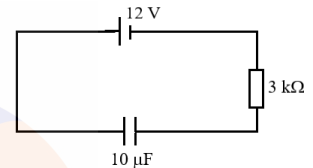
$$q = q_0 \left(1 - e^{-\frac{t}{RC}}\right) \rightarrow 30 = 10 \times 12 \left(1 - e^{-\frac{t}{0.03}}\right) \rightarrow t = 0.03 \ln \frac{4}{3}$$

Current flowing through the circuit:

$$i = \frac{dq}{dt} = I_0 e^{-\frac{t}{RC}} = \frac{12}{3000} e^{-\frac{0.03 \ln \frac{4}{3}}{RC}} = 4 \times 10^{-3} e^{-\frac{0.03 \ln \frac{4}{3}}{0.03}} = 4 \times 10^{-3} \times \frac{3}{4} = 3 \text{ mA}$$

Power dissipated through the resistor:

$$P = i^2 R = 9 \times 10^{-6} \times 3 \times 10^3 = 27 \times 10^{-3} = \mathbf{27 \text{ mW}}$$



34. A hollow conducting sphere of radius 15 cm has a uniform surface charge density $+3.2 \mu\text{C}/\text{m}^2$. When a point charge q is placed at the center of the sphere, the electric field at 25 cm from the center just reverses in direction keeping the magnitude the same. Therefore, q is

- a) $+0.91 \mu\text{C}$ b) $-0.91 \mu\text{C}$ c) $+1.81 \mu\text{C}$ d) **$-1.81 \mu\text{C}$**

Sol Field intensity due to charged spherical shell at a point outside: $E_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{\sigma}{\epsilon_0} \left(\frac{R}{r}\right)^2$ (+ve)

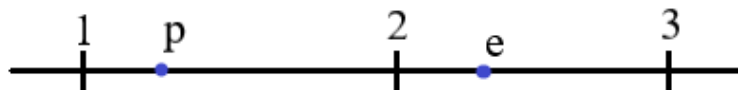
Field intensity due to point charge: $E_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$ (-ve)

Given: the field gets reversed with same magnitude: the point charge should be negative.

Given: $E = -E_1 \rightarrow -E_2 + E_1 = -E_1 \rightarrow E_2 = -2E_1$

$$\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = -2 \frac{\sigma}{\epsilon_0} \left(\frac{R}{r}\right)^2 \rightarrow q = -8\pi\sigma R^2 = 8 \times 3.14 \times 3.2 \times 10^{-6} \times 225 \times 10^{-4} = \mathbf{-1.81 \mu\text{C}}$$

35. An electron (e) and a proton (p) are situated on the straight line as shown below. The directions of the electric field at the points 1, 2 and 3 respectively, are shown as



- a) $\rightarrow \leftarrow \rightarrow$ b) $\leftarrow \rightarrow \leftarrow$ c) $\leftarrow \rightarrow \rightarrow$ d) $\rightarrow \leftarrow \leftarrow$

Sol Electric field at a point is directed away from +ve charge and towards – ve charge.

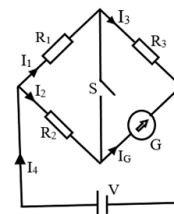
Point 1: field due to proton is strong and left, field due to electron is weak and right.

Point 2: field due to proton is weak and right, field due to electron is strong and right

Point 3: field due to proton is weak and right, field due to electron is strong and left

- 36.** In the circuit shown $R_1 \neq R_2$. The reading in the galvanometer is the same with switch S open or closed. Then,

- a) $I_1 = I_G$
- b) $I_2 = I_G$
- c) $I_3 = I_G$
- d) $I_4 = I_G$



Sol When the switch is open, current through R_2 and G is same: $I_2 = I_G$

When the switch is open, current through R_1 and R_3 is same: $I_1 = I_3$

When the switch is closed, it is given that the **reading of the galvanometer remains same.**

- 37.** A thin wire of length 1 m is placed perpendicular to the XY plane. If it is moved with velocity $\vec{v} = 4\hat{i} - \hat{j}$ m/s in the region of magnetic induction $\vec{B} = \hat{i} + 4\hat{j}$ wb/m². The potential difference developed between the ends of the wire is

- a) zero
- b) 3 V
- c) 15 V
- d) **17 V**

Sol Length of the wire: $\vec{l} = 1 \hat{k}$

Velocity of the wire: $\vec{v} = 4\hat{i} - \hat{j}$ m/s

Magnetic field induction: $\vec{B} = \hat{i} + 4\hat{j}$ wb/m²

Potential difference developed between the ends of the wire: $\Delta V = (\vec{v} \times \vec{B}) \cdot \vec{l}$

$\Delta V = 17 \hat{k} \cdot 1 \hat{k} = \mathbf{17 \text{ volt}}$

- 38.** A steel cooking pan has copper coating at its bottom. The thickness of copper coating is half the thickness of steel bottom. The conductivity of copper is three times that of steel. If the temperature of blue flame is 119 °C and that of the interior of the cooking pan is 91 °C, then the temperature at the interface between the steel bottom and the copper coating in steady state is

- a) 98 °C
- b) 103 °C
- c) **115 °C**
- d) 108 °C

Sol Thickness of copper coating = $\frac{1}{2}$ thickness of steel $\rightarrow l_{cu} = \frac{1}{2} l_{st}$

Conductivity of copper = 3 conductivity of steel $\rightarrow K_{cu} = 3 K_{st}$

Temperature at copper surface: 119 °C

Temperature at steel surface: 91 °C

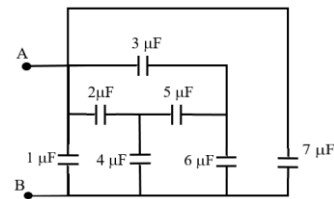
Temperature at the interface: θ

For steady state heat transfer: $\frac{dQ}{dt}$ is constant: $\frac{dQ}{dt} = \frac{KA\Delta\theta}{l}$

$$\frac{K_{cu}A_{cu}(119-\theta)}{l_{cu}} = \frac{K_{st}A_{st}(\theta-91)}{l_{st}} \rightarrow \frac{3K_{st}A_{cu}(119-\theta)}{0.5l_{st}} = \frac{K_{st}A_{st}(\theta-91)}{l_{st}} \rightarrow 6(119-\theta) = (\theta-91) \rightarrow \theta = \mathbf{115^\circ C}$$

39. The total capacitance between points A and B in the arrangement shown below is

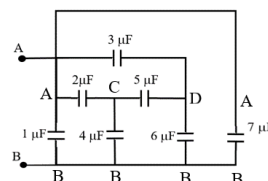
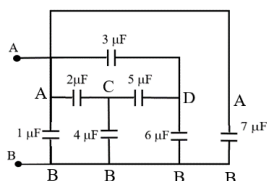
- a) $28 \mu F$ b) $\frac{34}{7} \mu F$
 b) $23 \mu F$ d) $\frac{34}{3} \mu F$



Sol ABCD forms a balanced Wheatstone's bridge.

Effective capacitance of the bridge: $\frac{2 \times 4}{2+4} + \frac{3 \times 6}{3+6} = \frac{4}{3} + 2 = \frac{10}{3} \mu F$

Effective capacitance between A & B: $\frac{10}{3} + 7 + 1 = \frac{34}{3} \mu F$



40. A fiber sheet of thickness 1 mm and a mica sheet of thickness 2 mm are introduced between two metallic parallel plates to form a capacitor. Given that the dielectric strength of fiber is 6400 kV/m and the dielectric constants of fiber and mica are 2.5 and 8 respectively, the electric field inside the mica sheet just at the breakdown of the fiber will be

2.5	8
fiber	mica
1 mm	2 mm

- a) **2000 kV/m** b) 2048 kV/m c) 3200 kV/m d) 6400 kV/m

Sol Let σ be the surface charge density.

Electric field intensity inside fiber sheet: $E_f = \frac{\sigma}{K_f \epsilon_0}$ ---- (1)

Electric field intensity inside mica sheet: $E_m = \frac{\sigma}{K_m \epsilon_0}$ ---- (2)

From (1) and (2): $\frac{E_f}{E_m} = \frac{K_m}{K_f} = \frac{8}{2.5} = 3.2 \rightarrow E_m = \frac{E_f}{3.2} = \frac{6400}{3.2} = \mathbf{2000 \text{ kV/m}}$

41. The position vector of a point mass is expressed as $\vec{r} = at\hat{i} + bt^2\hat{j}$. The trajectory of the particle is

- a) a straight line b) **a parabola** c) a hyperbola d) none of the above

Sol Position vector of the point mass: $\vec{r} = at\hat{i} + bt^2\hat{j}$

Position vector of a point mass: $\vec{r} = x\hat{i} + y\hat{j}$

X component of the position vector: $x = at$

Y component of the position vector: $y = bt^2$

Eliminating t from both the components: $y = b \left(\frac{x}{a}\right)^2 = \frac{b}{a^2} x^2$ ($y = kx^2$: equation of parabola)

42. In a series LCR circuit fed with an alternating emf $E = E_0 \sin \omega t$,

- a) the voltage across L is in phase with the applied emf E
 b) the voltage across C is in phase with the applied emf E
c) the voltage across R is in phase with the applied emf E
 d) the voltages across L, C and R are all in phase with the applied emf E

Sol Supply voltage: $E = E_0 \sin \omega t$

Current through the circuit: $i = \frac{v}{z} = I_0 \sin (\omega t \pm \phi)$

If $X_L > X_C$ current lags the applied emf: $i = I_0 \sin (\omega t - \phi)$

If $X_L < X_C$ current leads the applied emf: $i = I_0 \sin (\omega t + \phi)$

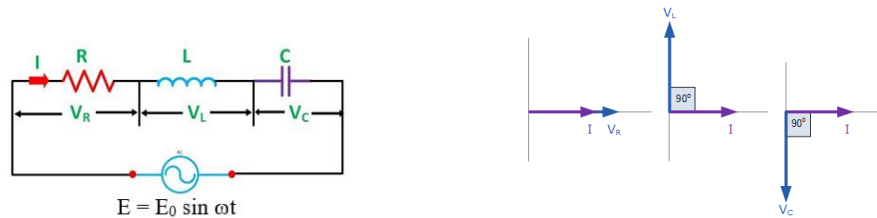
If $X_L = X_C$ (resonance) current is in phase with the applied emf.

Voltage across resistor is in phase with current

Voltage across inductor leads the current by 90°

Voltage across capacitor lags the current by 90°

Voltage across resistor is in phase with the applied emf at resonance.



43. A conducting wire is bent in the form of a n sided regular polygon enclosed by a circle of radius R . The magnetic field produced at its center by a current i flowing through the wire is

a) $\frac{\mu_0 i}{2R} \frac{\sin \frac{\pi}{n}}{\frac{\pi}{n}}$

b) $\frac{\mu_0 i}{2R} \frac{\cos \frac{\pi}{n}}{\frac{\pi}{n}}$

c) $\frac{\mu_0 i}{2R} \frac{\tan \frac{\pi}{n}}{\frac{\pi}{n}}$

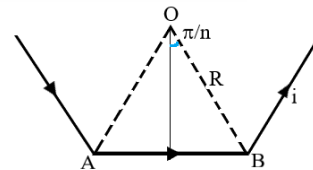
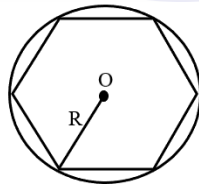
d) $\frac{\mu_0 i}{2R} \frac{\cot \frac{\pi}{n}}{\frac{\pi}{n}}$

Sol Perpendicular distance of point O from the line AB: $d = R \cos \frac{\pi}{n}$

Magnetic field due to AB at point O: $B_{AB} = \frac{\mu_0 i}{4\pi d} (\sin \alpha + \sin \beta) = \frac{\mu_0 i}{4\pi R \cos \frac{\pi}{n}} \left(2 \sin \frac{\pi}{n} \right)$

Magnetic field due to the entire polygon at point O: $B = n B_{AB}$

$B = n \frac{\mu_0 i}{4\pi R \cos \frac{\pi}{n}} \left(2 \sin \frac{\pi}{n} \right) = \frac{\mu_0 i}{2R} \frac{\tan \frac{\pi}{n}}{\frac{\pi}{n}}$



44. The effective resistance between points A and B in the circuit arrangement shown below is

a) **14 Ω**

b) 15 Ω

c) 30 Ω

d) none of the above

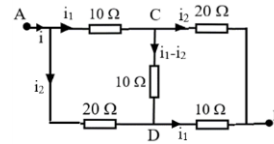
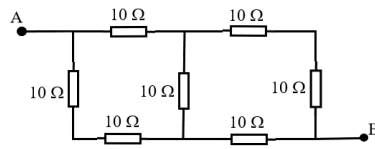
Sol Apply KVL through ACB: $v_A - 10i_1 - 20i_2 - v_B = 0 \rightarrow v_A - v_B = 10i_1 + 20i_2$ ---- (1)

Apply KVL through ACDB: $v_A - 10i_1 - 10(i_1 - i_2) - 10i_1 = 0 \rightarrow v_A - v_B = 30i_1 - 10i_2$ ---- (2)

From (1) & (2): $10i_1 + 20i_2 = 30i_1 - 10i_2 \rightarrow 2i_1 = 3i_2$

Effective resistance of the network between A & B: $R = \frac{v_{AB}}{i}$

$R = \frac{10i_1 + 20i_2}{i_1 + i_2} = \frac{10 \times 1.5i_2 + 20i_2}{1.5i_2 + i_2} = \frac{350}{25} = \mathbf{14 \Omega}$



45. The magnetic dipole moment of an electron in the S state of hydrogen atom revolving in a circular orbit of radius 0.0527 nm with a speed 2.2×10^6 m/s is (in Am^2)

a) 4.64×10^{-24} b) **9.28×10^{-24}** c) 18.56×10^{-24} d) 2.32×10^{-24}

Sol Magnetic dipole moment of a charged particle: $M = iA = \frac{q}{T} \pi r^2 = \frac{qv}{2\pi r} \pi r^2 = \frac{qvr}{2}$

$$M = \frac{1.6 \times 10^{-19} \times 2.2 \times 10^6 \times 0.0527 \times 10^{-9}}{2} = \mathbf{9.28 \times 10^{-24} \text{ Am}^2}$$

46. A steel cable hanging vertically can support a maximum load W . The cable is cut to exactly half of its original length, the maximum load that it can support now is

a) **W** b) $\frac{W}{2}$ c) $2W$ d) more than $\frac{W}{2}$ but less than W

Sol Young's modulus: $Y = \frac{Fl}{Ae} = \frac{Wl}{Ae}$

Maximum load supported by a cable is decided by the breaking stress of the cable.

Breaking stress: $\frac{F_{\max}}{A} \rightarrow F_{\max} = \text{Breaking stress (A)}$

$F_{\max} \propto A \rightarrow$ maximum load is **independent of length of the cable.**

47. The strings AB and AC each of length 40 cm, connect a ball of mass 200 g to a vertical shaft as shown. When the shaft rotates at a constant angular speed ω , the ball travels in a horizontal circle with the strings inclined at $\gamma = 30^\circ$ to the shaft. If the tension in the string AC is 4 N, that in the string AB and the angular speed ω respectively are

a) **6.26 N and 11.32 rad/s** b) 7.92 N and 14.32 rad/s
c) 7.92 N and 11.32 rad/s d) 6.26 N and 14.32 rad/s

Sol Lengths of the strings AB & AC: 40 cm = 0.4 m

Mass of the ball: $m = 200 \text{ g} = 0.2 \text{ kg}$

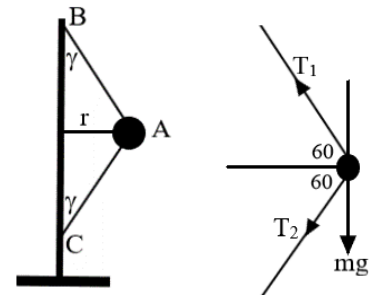
Radius of the circular path: $\sin \gamma = \frac{r}{l} \rightarrow \sin 30 = \frac{r}{0.4} \rightarrow r = 0.2 \text{ m}$

For vertical equilibrium:

$$T_1 \sin 60 = mg + T_2 \sin 60 \rightarrow \frac{\sqrt{3}}{2} T_1 = 1.96 + 2\sqrt{3} \rightarrow \mathbf{T_1 = 6.26 \text{ N}}$$

For horizontal equilibrium: $(T_1 + T_2) \cos 60 = m\omega^2 r$

$$(6.26 + 4) 0.5 = 0.2 \times 0.2 \times \omega^2 \rightarrow \omega^2 = 128.25 \rightarrow \mathbf{\omega = 11.32 \text{ rad/s}}$$



48. tightly wound long solenoid carries a current 5 A. An electron shot perpendicular to the solenoid axis inside it revolves at a frequency 10^8 rev/s. The number of turns per meter length of the solenoid

a) 57 b) 176 c) **569** d) 352

Sol Current carried by the solenoid: $i = 5$ A

Angular frequency of the electron: 10^8 rev/s = $2\pi \times 10^8$ rad/s

Magnetic field of a solenoid: $B = \mu_0 i n$

Since the electron is projected perpendicular to the field, it will follow circular path.

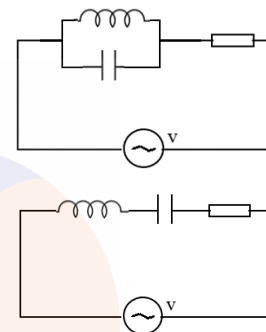
The centripetal force required for circular path is provided by magnetic force: $F_m = Bev = \mu_0 i n e v$

$$\text{Centripetal force: } F_c = \frac{mv^2}{r} = \mu_0 i n e v \rightarrow n = \frac{mv}{\mu_0 i e r} = \frac{\omega}{\mu_0 i \frac{e}{m}}$$

$$n = \frac{2\pi \times 10^8}{4\pi \times 10^{-7} \times 5 \times 1.76 \times 10^{11}} = \cong \mathbf{569}$$

49. The same alternating voltage $v = V_0 \sin(\omega t)$ is applied in both the LCR circuits shown below. The current through the resistance R at resonance is

a) maximum in fig. (1) and maximum in fig. (2)
 b) minimum in fig. (1) and maximum in fig. (2)
 c) **maximum in fig. (1) and minimum in fig. (2)**
 d) minimum in fig. (1) and minimum in fig. (2)



Sol Alternating voltage supplied to the circuit: $v = V_0 \sin(\omega t)$

Impedance of series RLC network: $z = \sqrt{R^2 + (X_L - X_C)^2}$

Series resonance: Impedance of the circuit is minimum ($z = R$),

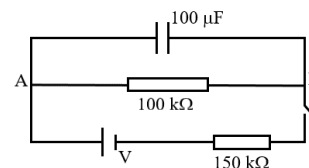
hence current is maximum ($X_L = X_C$)

Impedance of parallel LC network: $z = \frac{1}{\sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_L} - \frac{1}{X_C}\right)^2}}$

Parallel resonance: Impedance of the circuit is maximum, hence current is minimum.

50. The switch S in the circuit shown is closed for a long time and then opened at time $t = 0$. The current in the $100 \text{ k}\Omega$ resistance at $t = 3 \text{ s}$ is

a) zero b) $10 \mu\text{A}$ c) $35.5 \mu\text{A}$ d) $16 \mu\text{A}$



Sol Steady state current through the battery: $I_0 = \frac{V}{250} \text{ mA}$

Steady state charge attained by the capacitor:

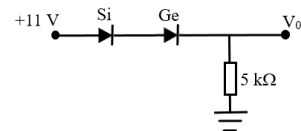
$$q_0 = CV_{AB} \rightarrow q_0 = 100 \times 10^{-6} \times \frac{V}{250} \times 10^{-3} \times 100 \times 10^3 = \frac{V}{25} \text{ mC}$$

When the switch is opened, the capacitor discharges through $100 \text{ k}\Omega$ resistor.

The battery voltage is required to get a numerical answer.

51. In the network shown below the voltage V_0 is nearly

- a) **10 volt** b) 11volt c) 12 volt d) Zero



Sol Threshold voltage: the forward voltage at which the current in the diode increases exponentially.

For silicon: $v_{Si} = 0.7$ volt

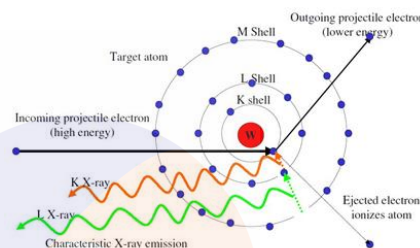
For Germanium: $v_{Ge} = 0.3$ volt

Applying KVL between input and output: $11 - 0.7 - 0.3 - V_0 = 0 \rightarrow V_0 = 10$ **volt**

52. The energy of the characteristic X ray photon in a Coolidge tube comes from

- a) the kinetic energy of striking electron
b) the kinetic energy of the free electrons of the target
c) the kinetic energy of the ions of the target
d) **the electronic transition of the target atom**

Sol Characteristic X rays are formed when a fastmoving electron from the cathode dislodges an inner electron from the target atom and an electron in the higher energy levels of the target atom jumps down to fill the gap. So, the energy of characteristic X ray photon is nothing, but **the transition energy** released in the target atom.



53. The maximum wavelength that can ionize a hydrogen atom initially in the ground state is

- a) 660.0 nm b) 364.5 nm c) 121.9 nm d) **91.4 nm**

Sol Energy of the hydrogen in ground state: $E = 13.6$ eV

Wavelength of the radiation: $E = \frac{hc}{\lambda} \rightarrow \lambda = \frac{hc}{E} = \frac{1240 \text{ eV nm}}{13.6 \text{ eV}} = 91.4$ **nm**

54. The wavelength of the waves associated with a proton and a photon are the same. Therefore, the two have equal

- a) mass b) velocity c) **momentum** d) kinetic energy

Sol Wavelength associated with a moving proton: $\lambda_{pr} = \frac{h}{p_{pr}}$

Wavelength associated with a photon: $\lambda_{ph} = \frac{h}{p_{ph}}$

Given: $\lambda_{pr} = \lambda_{ph} \rightarrow \frac{h}{p_{pr}} = \frac{h}{p_{ph}} \rightarrow p_{pr} = p_{ph}$

55. Which of the following sources emits light having highest degree of coherence?

- a) LED b) **LASER diode** c) neon lamp d) incandescent lamp

Sol LASER: Light Amplification by Stimulated Emission of Radiation

Monochromacy: The wavelength range of LASER is very low. So, we obtain almost single color (monochromatic) light.

Directionality: LASER travels very long distances without much deviation from its original direction (path).

High Coherence: High degree of coherence is possible for LASER as it is produced by stimulated emission.

High Intensity: It is possible to concentrate high energy on a small area using LASER beams.

56. An alpha particle with kinetic energy K approaches a stationary nucleus having atomic number z . The distance of closest approach is b . Therefore, the distance of closest approach for a nucleus of atomic number $2z$ is

- a) $b/2$ b) $\sqrt{2}b$ c) **$2b$** d) $4b$

Sol Initial kinetic energy of the alpha particle: K

Atomic number of the stationary nucleus: z

Conservation of energy: initial KE = final PE

$$K = \frac{1}{4\pi\epsilon_0} \frac{(2e)(ze)}{b} \text{ here } b \text{ is distance of closest approach at which the alpha particle stops.}$$

$$K = \frac{1}{4\pi\epsilon_0} \frac{(2e)(2ze)}{B} \text{ here } B \text{ is the distance of closest approach at which the alpha particle stops.}$$

Comparing both the expressions: **$B = 2b$**

57. In a photodiode the reverse current increases when exposed to light of wavelength 620 nm or less. The band gap of the semiconductor used is

- a) 0.67 eV b) 1.12 eV c) **2.00 eV** d) 2.42 eV

Sol Wavelength required to increase the reverse current: $\lambda = 620 \text{ nm}$

$$\text{Band gap of the semiconductor: } E_G = \frac{hc}{\lambda} = \frac{1240}{620} = \textbf{2.00 eV}$$

58. An electron in hydrogen atom jumps from a level $n = 4$ to $n = 1$. The momentum of the recoiled atom is

- a) **$6.8 \times 10^{-27} \text{ kg-m/s}$** b) $12.75 \times 10^{-19} \text{ kg-m/s}$
c) $13.6 \times 10^{-19} \text{ kg-m/s}$ d) zero

Sol Transition of the electron in hydrogen atom: $n = 4$ to $n = 1$

$$\text{Energy released during this transition: } E = -13.6 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

$$E = 13.6 \left(1 - \frac{1}{16} \right) = 13.6 \times \frac{15}{16} = 12.75 \text{ eV} = 12.75 \times 1.6 \times 10^{-19} = 20.4 \times 10^{-19} \text{ J}$$

$$\text{Momentum of the electron: } P = \frac{E}{c} = \frac{20.4 \times 10^{-19}}{3 \times 10^8} = \textbf{6.8 \times 10^{-27} kg m/s}$$

59. For the Boolean equation $Y = AB + A(B + C) + B(B + C) + \bar{B}$, which of the following statements is correct?

- a) Y does not depend on A but depends on B b) Y does not depend on B but depends on A
c) **Y does not depend on B** d) Y depends only on C

Sol Given Boolean equation: $Y = AB + A(B + C) + B(B + C) + \bar{B}$

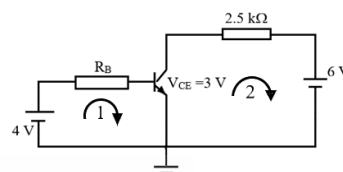
$$Y = AB + AB + AC + BB + BC + \bar{B}$$

$$Y = AB + AC + BC + 1$$

Irrespective of the values of A, B and C, the output is always 1.

So, the **output is independent of all the three input variables.**

60. Refer to the common emitter amplifier circuit shown below, using a transistor with $\beta = 80$ and $V_{BE} = 0.7$ V. The value of resistance R_B is



- a) 330Ω b) $330 \text{ k} \Omega$ c) 220Ω d) **$220 \text{ k} \Omega$**

Sol Apply KVL to first loop: $4 - i_B R_B - V_{BE} = 0 \rightarrow i_B R_B = 3.3 \text{ volt} \text{ ---- (1)}$

Apply KVL to second loop: $-i_C R_C - 6 + V_{CE} = 0 \rightarrow i_C R_C = 3 \rightarrow i_C = \frac{3}{2.5 \times 10^3} = 1.2 \text{ mA}$

Current amplification factor: $\beta = \frac{i_C}{i_B} \rightarrow 80 = \frac{1.2}{i_B} \rightarrow i_B = 15 \mu\text{A}$

From (1): $i_B R_B = 3.3 \rightarrow 15 \times 10^{-6} R_B = 3.3 \rightarrow \mathbf{R_B = 220 \text{ k}\Omega}$

61. A horizontal insulated cylinder of volume V is divided into four identical compartments by stationary semi-permeable thin partitions as shown. The four compartments from left are initially filled with 28 g helium, 160 g oxygen, 28 g nitrogen and 20 g hydrogen respectively. The left partition lets through hydrogen, nitrogen and helium while the right partition lets through hydrogen only. The middle partition lets through hydrogen and nitrogen both. The temperature T inside the entire cylinder is maintained constant. After the system is set in equilibrium

- a) **pressure of helium is $\frac{14RT}{V}$**
c) **pressure of nitrogen is $\frac{4RT}{3V}$**

- b) **pressure of oxygen is $\frac{20RT}{V}$**
d) **pressure of hydrogen is $\frac{10RT}{V}$**

Sol Volume of each part of the container: $V/4$

Pressure of helium: $P_{He} = \frac{n_{He}RT}{V/4} = \frac{14RT}{V}$

Pressure of oxygen: $P_O = \frac{n_O RT}{V/4} = \frac{20RT}{V}$

Pressure of nitrogen: $P_N = \frac{n_N RT}{3V/4} = \frac{4RT}{3V}$

Pressure of hydrogen: $P_H = \frac{n_H RT}{V} = \frac{10RT}{V}$

28 g He 7	160 g O ₂ 5	28 g N ₂ 1	20 g H ₂ 10
28RT/V	20RT/V	4RT/V	40RT/V

He N ₂ H ₂	He O ₂ N ₂ H ₂	N ₂ H ₂	H ₂
--	--	----------------------------------	----------------

62. After charging a capacitor C to a potential V, it is connected across an ideal inductor L. The capacitor starts discharging simple harmonically at time $t = 0$. The charge on the capacitor at a later time instant is q and the periodic time of simple harmonic oscillations is T . Therefore

a) $q = CV \sin(\omega t)$ b) $q = CV \cos(\omega t)$ c) $T = 2\pi \sqrt{\frac{1}{LC}}$ d) $T = 2\pi \sqrt{LC}$

Sol Initially, the capacitor is charged.

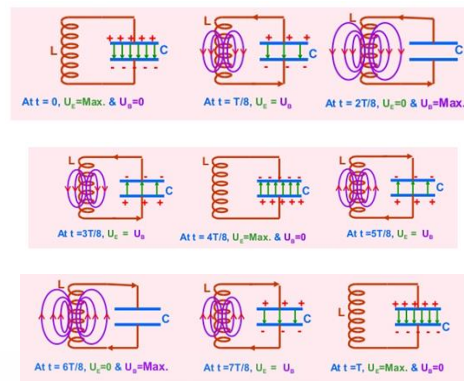
Apply KVL: $-\frac{q}{C} - L \frac{di}{dt} = 0$

$-\frac{q}{C} - L \frac{d^2q}{dt^2} = 0 \rightarrow \frac{d^2q}{dt^2} + \frac{q}{LC} = 0$ ($a + \omega^2 x = 0$)

$\omega = \frac{1}{\sqrt{LC}} \rightarrow \frac{2\pi}{T} = \frac{1}{\sqrt{LC}} \rightarrow T = 2\pi \sqrt{LC}$

The solution of the second order differential equation is of the form

$q = q_0 \cos(\omega t) = CV \cos(\omega t)$



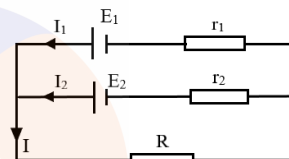
63. In the circuit arrangement shown two cells supply a current I to a load resistance $R = 9 \Omega$. One cell has an emf $E_1 = 9 \text{ V}$ and internal resistance $r_1 = 1 \Omega$ and the other cell has an emf $E_2 = 6 \text{ V}$ and internal resistance $r_2 = 3 \Omega$. The currents are as shown. Then,

a) $I_1 = 0.9 \text{ A}$ and $I_2 = 0.5 \text{ A}$

b) $I \cong 0.85 \text{ A}$

c) if the cell of emf E_1 is removed, current I will be smaller

d) if the cell of emf E_2 is removed, current I will be smaller



Sol

Effective emf of the parallel combination of the cells: $e = \frac{\sum(E/r)}{\sum(1/r)} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\frac{9}{1} + \frac{6}{3}}{\frac{1}{1} + \frac{1}{3}} = \frac{33}{4} \text{ volt}$

Effective resistance of the circuit: $r = R + \frac{r_1 r_2}{r_1 + r_2} = 9 + \frac{1 \times 3}{1 + 3} = \frac{39}{4} \Omega$

Current through the external resistor: $I = \frac{e}{r} = \frac{33/4}{39/4} \cong 0.85 \text{ A}$

When the cell of emf E_1 is removed: $I = \frac{E_2}{R + r_2} = \frac{6}{9 + 3} = 0.5 \text{ A}$

When the cell of emf E_2 is removed: $I = \frac{E_1}{R + r_1} = \frac{9}{9 + 1} = 0.9 \text{ A}$

64. A transparent cylindrical rod of length $l = 50 \text{ cm}$, radius $R = 10 \text{ cm}$ and refractive index $\mu = \sqrt{3}$ lies onto a horizontal plane surface. A ray of light moving perpendicular to its length is incident on the rod horizontally at a height h above the plane surface such that this ray emerges out of the rod at a height 10 cm above the plane surface. Therefore, h is

a) 1.34 cm

b) 1.73 cm

c) 10.0 cm

d) 18.66 cm

Sol From the fig: $i = 2r$

(external angle in $\triangle ACD$ = sum of opposite internal angles)

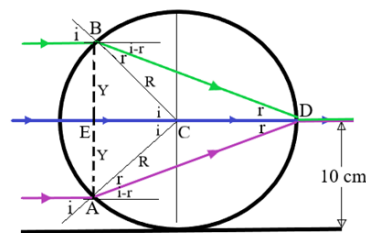
Snell's law at A: $n_1 \sin \theta_1 = n_2 \sin \theta_2 \rightarrow 1 \times \sin i = \sqrt{3} \sin r$

$\sin 2r = \sqrt{3} \sin r \rightarrow r = 30^\circ$ and $i = 60^\circ$

From $\triangle ACE$: $\sin i = \frac{AE}{AC} \rightarrow \sin 60 = \frac{Y}{R} \rightarrow Y = 10 \times \frac{\sqrt{3}}{2} = 5\sqrt{3}$ cm

Height of point A: $10 - 5\sqrt{3} = \mathbf{1.34}$ cm

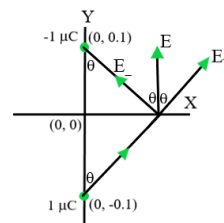
Height of point B: $10 + 5\sqrt{3} = \mathbf{18.66}$ cm



65. Two-point charges $+1 \mu\text{C}$ and $-1 \mu\text{C}$ are placed at points $(0, -0.1 \text{ m})$ and $(0, +0.1 \text{ m})$ respectively in XY plane. Then, choose the correct statement/s from the following

- a) The electric field at all points on the Y axis has the same direction
b) The dipole moment of $0.2 \mu\text{C} - \text{m}$ along +ve X direction

- c) **No work has to be done in bringing a test charge from infinity to the origin**
d) **Electric field at all points on the X axis is along +ve Y axis**



Sol The given arrangement forms an electric dipole.

Electric field is directed away from +ve charge and towards -ve charge.

Potential due to the dipole at the midpoint: $v_0 = \frac{kq}{r} - \frac{kq}{r} = 0$

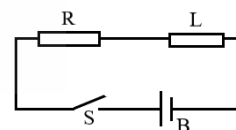
So, the work done in bringing a test charge from infinity to the origin: **$w = v_0 q = 0$**

Electric field at any point on the x axis: $E_+ = E_- = \frac{kq}{a^2}$

Since the two fields (vectors) are equal in magnitude, their resultant is angular bisector.

So, **the electric field at all points on the X axis is along +ve Y direction**

66. An inductance L, a resistance R and a battery B are connected in series with a switch S. The voltages across L and R are V_L and V_R respectively. Just after closing the switch S,



- a) **V_L will be greater than V_R**
b) V_L will be less than V_R
c) V_L will be same as V_R
d) **V_L will decrease while V_R will increase as time progresses**

Sol Current through series RL network: $i = I_0 \left(1 - e^{-\frac{t}{\tau}}\right)$

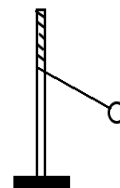
Voltage across resistor: $V_R = iR = I_0 R \left(1 - e^{-\frac{t}{\tau}}\right) = V_0 \left(1 - e^{-\frac{t}{\tau}}\right)$

Voltage across inductor: $V_L = V_0 - V_R = V_0 e^{-\frac{t}{\tau}}$

Just after closing the switch: $t = 0$; $V_R = 0$ and $V_L = V_0$

As the time passes, **V_L will decrease while V_R will increase.**

67. A string of length l , tied to the top of a pole, carries a ball at its other end as shown. On giving the ball a single hand blow perpendicular to the string, it acquires an initial velocity v_0 in the horizontal plane and moves in a spiral of decreasing radius by curling itself around the pole. Therefore,



a) the instantaneous center of revolution of the ball is the point of contact of the string with the pole at that instant

b) the instantaneous center of revolution of the ball will be fixed at the point where the string was initially fixed

c) the angular momentum of the system will not be conserved

d) the angular momentum of the system will be conserved

- Sol**
- (a) The point of contact of the string with the pole is at rest at any point of time. So, the instantaneous center of revolution of the ball is the point of contact of the string with the pole at that instant.
- (c) Weight of the ball (gravitational force = mg) acts vertically downward. The torque provided by this force about the pole is non-zero. So, the angular momentum of the system is not conserved

68. A circular loop of conducting wire of radius 1 cm is cut at a point A on its circumference. It is then folded along a diameter through A such that two semicircular loops lie in two mutually perpendicular planes. In this region a uniform magnetic field B of magnitude 100 mT is directed perpendicular to the diameter through A and makes angles 30° and 60° with the planes of the two semicircles. The magnetic field reduces at a uniform rate from 100 mT to zero in a time interval of 4.28 ms. Therefore,

a) instantaneous emfs in the two loops are in the ratio $\sqrt{2} : 1$

b) instantaneous emfs in the two loops are in the ratio $\sqrt{3} : 1$

c) the total emf between free ends at point A is 5 mV

d) the total emf between free ends at point A is 1.4 mV

- Sol** Flux of magnetic field through a coil: $\phi = BA \cos \theta$

Magnitude of induced emf in the coil: $e = \frac{d\phi}{dt} = A \cos \theta \frac{dB}{dt} \rightarrow e \propto \cos \theta$

Ratio of emf induced in the semicircular coils: $\frac{e_1}{e_2} = \frac{\cos \theta_1}{\cos \theta_2} = \frac{\cos 30}{\cos 60} = \sqrt{3}$

Total emf between the free ends at point A: $e = e_1 - e_2 = A \frac{dB}{dt} (\cos \theta_1 + \cos \theta_2)$

$$e = \frac{\pi(0.01)^2}{2} \frac{100}{4.28} \left(\frac{\sqrt{3}}{2} - \frac{1}{2} \right) = \frac{25\pi}{4.28} \times 0.732 \times 10^{-4} = 1.4 \text{ mV}$$

69. A converging lens of focal length 40 cm is fixed at 40 cm in front of a screen. An object placed 120 cm from the fixed lens is required to be focused on the screen by introducing another identical lens in between. The second lens should be placed at a distance x from the object where x is
- a) 40 cm b) 50 cm c) 140 cm d) 150 cm

Sol Let the second lens is placed to the left of first lens.
 Screen is placed at the focus of the 1st lens. To catch the img. on the screen, obj. should be at infinity.
 Image formed by 2nd lens acts as object for 1st lens. So, the 2nd lens should form image at infinity.
 Lens formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \rightarrow \frac{1}{\infty} + \frac{1}{x} = \frac{1}{40} \rightarrow x = 40 \text{ cm}$
 Let the second lens be placed to the right of first lens.
 Lens formula for first lens: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \rightarrow \frac{1}{v} + \frac{1}{120} = \frac{1}{40} \rightarrow v = 60 \text{ cm}$
 Let the second lens be placed at a distance y from the screen.
 Lens formula for second lens: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \rightarrow \frac{1}{y} - \frac{1}{y+20} = \frac{1}{40} \rightarrow y = 20 \text{ cm}$
 So, the distance of second lens from the object: $x = 120 + 40 - 20 = 140 \text{ cm}$



70. Mysteriously, a charged particle moving with velocity $\vec{v} = v_0 \hat{i}$ entered the tube of Thomson's apparatus where the parallel metallic plates of length 5 cm along X axis are separated by 2 cm. Under the influence of a magnetic field $\vec{B} = 4.57 \times 10^{-2} \hat{k} \text{ T}$, the particle is found to deflect by an angle of 5.7° . When a potential of 2000 V is applied between the two plates, the particle is found to move straight to the screen without any deflection. Then

- a) the velocity $v_0 = 2.19 \times 10^6 \text{ m/s}$
- b) the charge to mass ratio of the particle is $9.58 \times 10^7 \text{ C/kg}$
- c) radius of the circular path in the magnetic field is 50 cm
- d) the particle is identified as a proton

Sol Velocity of the particle: $v_0 = \frac{E}{B} = \frac{\Delta V}{d(B)} = \frac{2000}{0.02 \times 4.57 \times 10^{-2}} = 2.19 \times 10^6 \text{ m/s}$

Deviation in the path of the particle: $\sin \theta = \frac{l}{r} \rightarrow r = \frac{5}{0.1} = 50 \text{ cm}$

Radius of the circular path followed by charged particle in transverse magnetic field:

$$r = \frac{mv}{Bq} = \frac{v}{B \frac{q}{m}} \rightarrow \frac{q}{m} = \frac{v}{B r} = \frac{2.19 \times 10^6}{4.57 \times 10^{-2} \times 0.5} = 9.58 \times 10^7 \text{ C/kg}$$

$$\frac{q}{m} = 9.58 \times 10^7 \rightarrow \frac{1.6 \times 10^{-19}}{m} = 9.58 \times 10^7 \rightarrow m = 1.67 \times 10^{-27} \text{ kg (proton)}$$